

Is This Relation a Function?

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|---|---|
| 1. (a) output paired with 3 = 9, — | 6. (a) output when the input is 7 = 15, (b) |
| 2. (a) output paired with 6 = 11, (b) the two | output when the input is 12 = 25 |
| outputs of the repeated input = <, — | 7. (a) the two y-values compared = <, — |
| 3. (a) y-value at that point = 7, — | 8. (a) output paired with 3 = 4, — |
| 4. output when the input is 5 = 17 | 9. (a) output of R at 4 = 12, (b) the two |
| 5. (a) output paired with 9 = 5, (b) the two | outputs of the repeated input in S = <, — |
| outputs of the repeated input = >, — | 10. (a) output when the input is 6 = 23, — |
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What is verified

14 of 22 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 1, 2, 3, 5, 7, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.F.A.1 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–4 of 5 across 22 tagged checks.

1. Table: inputs 1, 2, 3, 4 against outputs 5, 7, 9, 11. (a) Output paired with 3. (b) Is it a function?

Solution (a): Read down from the input 3.

9

Solution (b) — **instructor-judged.** It is a function. Every input in the top row appears exactly once, so no input is left with a choice of outputs. Saying that no input is repeated is the same reason and earns full credit; a bare “yes” does not.

2. Pairs (5, 5), (4, 9), (5, 8), (6, 11). (a) Output paired with 6. (b) Compare the two outputs of the repeated input. (c) Is it a function?

Solution (a): The pair beginning with 6 is (6, 11).

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Solution (b): The input 5 appears in (5, 5) and (5, 8), so its two outputs are 5 and 8, in that order.

5 < 8

Solution (c) — **instructor-judged.** It is not a function. The input 5 is paired with 5 in one pair and with 8 in another, and a function allows each input only one output. Naming the offending input and both of its outputs is the full answer.

3. Graph P. (a) y -value of the point at $x = 4$. (b) Is it a function?

Solution (a): The plotted point four across sits seven up.

7

Solution (b) — instructor-judged. It is a function. No two plotted points sit directly above each other, so no input value carries more than one output — the vertical-line test: any vertical line meets the plotted points at most once. “Yes” with no test named is not enough.

4. Rule $y = 4x - 3$. Output when the input is 5.

Solution: Substitute 5 for x : $4(5) - 3 = 20 - 3$.

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5. Table: inputs 8, 6, 8, 9 against outputs 12, 5, 8, 5. (a) Output paired with 9. (b) Compare the two outputs of the repeated input. (c) Is it a function?

Solution (a): Read down from the input 9.

5

Solution (b): The input 8 appears in the first and third columns, with outputs 12 and 8 in that order.

12 > 8

Solution (c) — instructor-judged. It is not a function. The input 8 appears twice with two different outputs, 12 and 8, so that input has no single output. Note that the repeated *output* 5, which appears under inputs 6 and 9, is entirely allowed — the rule restricts inputs only.

6. Rule $y = 2x + 1$. (a) Output at input 7. (b) Output at input 12.

Solution (a): $2(7) + 1 = 14 + 1$.

15

Solution (b): $2(12) + 1 = 24 + 1$.

25

7. Graph Q. (a) Compare the y -values of the two stacked points. (b) Is it a function?

Solution (a): The two points directly above each other are at heights 1 and 6, lower first.

1 < 6

Solution (b) — instructor-judged. It is not a function. Two plotted points sit directly above each other, so one input value is paired with two different outputs — a vertical line through them meets the graph twice. A bare “no” does not earn credit.

8. Pairs (1, 4), (2, 4), (3, 4), (4, 4). (a) Output paired with 3. (b) Is it a function even though the output repeats?

Solution (a): The pair beginning with 3 is (3, 4).

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Solution (b) — instructor-judged. It is a function. The definition restricts inputs, not outputs: each of the inputs 1, 2, 3 and 4 appears once and has exactly one output. Several inputs sharing the output 4 is perfectly allowed — this is the point of the problem, and “no, because the outputs repeat” is the misconception to correct.

9. R : inputs 1, 2, 3, 4 against outputs 6, 8, 10, 12. S : pairs (4, 3), (6, 12), (4, 11), (8, 12). (a) R at input 4. (b) Compare the two outputs of the repeated input in S . (c) Which is a function?

Solution (a): Read down from the input 4 in R ’s table.

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Solution (b): In S the input 4 appears in (4, 3) and (4, 11), so its outputs are 3 and 11 in that order.

3 < 11

Solution (c) — instructor-judged. R is a function: each of its inputs appears once, so each has exactly one output. S is not: the input 4 appears twice with the different outputs 3 and 11. Note that S also repeats the output 12, under the inputs 6 and 8, and that on its own would have been fine. A correct answer checks each relation against the one-output-per-input rule.

10. Rule $y = 3x + 5$. (a) Output at input 6. (b) Why a rule always gives a function.

Solution (a): $3(6) + 5 = 18 + 5$.

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Solution (b) — instructor-judged. A rule performs the same arithmetic on whatever input it is given, and that calculation has only one possible result, so an input can never come back with two different outputs. A list of pairs is written out by hand and nothing stops the same input being written twice with different partners — which is exactly what happened in problems 2, 5 and 9. A restatement of the definition with no comparison of the two forms does not earn full credit.